Donghyun Son

dhson@stanford.edu / Google Scholar / Personal Website / X

Education

Stanford University, *M.S. in Computer Science* **Sep 2026 — Jun 2028 (expected)**

Seoul National University, *B.S. in Computer Science* **Mar 2018 — Feb 2026**

***GPA**: 4.04/4.3 (4.11/4.3 in major), **Summa Cum Laude**, includes 3 years of mandatory military service*

University of Texas at Austin, *Exchange Student, Computer Science* **Sep 2025 — Dec 2025**

Publications

*: equal contribution.

Preprints

[1] *Tailoring the Quantization Space for 1-Bit KV Cache Compression*
Minsoo Cheong*, **Donghyun Son***, Sungjoo Yoo

[2] NALAR*: Workflow-Aware Management of Agentic Applications*
Saurabh Agarwal*, Benedict Marco Laju*, **Donghyun Son**, Nitin Kedia, Myungjin Lee, Jayanth Srinivasa, Aditya Akella

Conference & Workshop Papers

[1] *EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models*
Minsoo Cheong, **Donghyun Son**, Woosang Lim, Sungjoo Yoo
NeurIPS 2026, AdaptFM@ICML 2026

[2] *NSNQuant: A Double Normalization Approach for Calibration-Free Low-Bit Vector Quantization of KV Cache*
Donghyun Son, Euntae Choi, Sungjoo Yoo
NeurIPS 2025

[3] *Gradient Estimation for Unseen Domain Risk Minimization with Pre-Trained Models*
Byounggyu Lew*, **Donghyun Son***, Buru Chang
OOD-CV@ICCV 2023

[4] *Reliable Decision from Multiple Subtasks through Threshold Optimization: Content Moderation in the Wild*
Donghyun Son*, Byounggyu Lew*, Kwanghee Choi*, Yongsu Baek, Seungwoo Choi, Beomjun Shin, Sungjoo Ha, Buru Chang
WSDM 2023 (Oral)

Technical Reports

[1] *K-EXAONE 2.0 Technical Report*
LG AI Research, contributed to inference infrastructure

Work Experience

Graduate Research Assistant, Stanford University **Oct 2026 — Present**

Ermon Group @ Stanford AI Lab (PI: Prof. Stefano Ermon) *Stanford, CA*

- Investigating efficient recurrent architectures for language models

Researcher, Seoul National University **Sep 2024 — Aug 2025 / Jul 2026 — Sep 2026**

CMALab @ SNU (Advisor: Prof. Sungjoo Yoo) *Seoul, South Korea*

- Investigated a calibration-free vector quantization method for KV cache, incorporating a novel normalization algorithm and efficient CUDA kernels for 1–2 bit inference (NeurIPS25)
- Proposed a vector quantization method for KV cache which tailors the quantization space to reflect error sensitivity and inter-channel dependencies (under review)

AI Scientist Intern, EXAONE Lab **Apr 2026 — Jul 2026**

LG AI Research (Mentor: Hyunjik Jo) *Seoul, South Korea*

- Integrated and optimized K-EXAONE and EXAONE 4.5 across LLM serving frameworks (vLLM, SGLang)
- Implemented EXAONE 4.5 model support for Ollama, including MTP-based speculative decoding
- Investigated reinforcement learning pipelines to identify sources of training–inference mismatch

Undergraduate Researcher, University of Texas at Austin **Aug 2025 — Dec 2025**

UTNS Research Group @ UT Austin (Mentor: Dr. Saurabh Agarwal, PI: Prof. Aditya Akella) *Austin, TX*

- Designed and implemented NALAR, a serving framework that dynamically schedules, migrates, and scales stateful agent workflows using workflow-level visibility and two-level control. (under review)

Machine Learning Engineer, Moderation ML Team **Mar 2022 — Jul 2023**

Hyperconnect (Acquired by Match Group) *Seoul, South Korea*

- Built an ML-based system handling 1M+ daily requests for moderating images and audio across Match Group brands (e.g., Tinder, Hinge)
- Devised a threshold optimization algorithm based on gradient descent to optimize the tradeoff between reliability and cost (WSDM23, Oral)
- Proposed a training algorithm for ML models to generalize effectively across different services and domains (OOD-CV@ICCV23)

Research Engineer, R&D Team **Aug 2020 — Mar 2022**

VisualCamp *Seoul, South Korea*

- Designed and implemented C++-based multi-threaded ML inference pipeline for on-device low-latency gaze estimation
- Built a training pipeline robust to data corruption and reduced MSE of a gaze estimation model by 30%

Honors & Awards

Competitions

- Master, at Codeforces (handle: **diordhd**, rating: 2200+)
- **2nd prize**, at Union of Clubs Programming Contest (UCPC) 2025
- **5th prize**, at Samsung Collegiate Programming Contest (SCPC) 2021
- **5th place**, at ICPC NERC Huawei Challenge 2020: Cloud Scheduling Challenge
- **Bronze Prize**, at 42nd Undergraduate Mathematics Contest Division 1 (for mathematical majors)

Scholarships

- **Study Abroad Scholarship of Kwanjeong Educational Foundation** ($60,000), from Kwanjeong Educational Foundation
- **Samsung Software Membership**, from Samsung Research (Nov 2021 — Feb 2026)
- **Korea-U.S. STEM Student Exchange Scholarship** ($9,000), from Ministry of Trade, Industry and Energy (2025.09)
- **Merit-based Full Tuition Academic Scholarship** ($2,300), from Seoul National University (2020-1, 2023-2, 2024-1, 2024-2)

Presentations

- **KV Cache Compression for Long Context Inference**, Weekly Seminar @ Deepest S16 (January 2025)
- **Efficient Algorithms for LLM Inference**, Weekly Seminar @ Deepest S15 (July 2024)

Additional Information

Technical Skills: C++, Python, Pytorch, Tensorflow, Jax, CUDA, OpenCL, MPI, OpenMP, vLLM, SGLang

Conference & Workshop Reviewer: NeurIPS 2026 (**Top Reviewer**), AdaptFM@ICML 2026, LCFM@NeurIPS 2026